

# Letters

## RESEARCH LETTER

### Trends in Hospitalizations and Liver Transplants Associated With Alcohol-Induced Liver Disease

Between 2015 and 2019, excessive alcohol consumption led to 1 in 5 deaths among US adults 20 to 49 years of age.<sup>1</sup> Liver transplant (LT) is the definitive treatment for decompensated alcohol-induced liver disease (AILD), which, since 2016, has become the foremost indication for LT.<sup>2</sup> There is a lack of studies examining demographic factors associated with AILD or AILD-related LT. Therefore, we investigated trends in hospitalizations and LTs associated with AILD in the US from 2005 to 2021.

**Methods** | The National Inpatient Sample (NIS; a publicly available all-payer inpatient care US database)<sup>3</sup> was used to analyze AILD-related hospitalizations using diagnosis codes 5710-5713 (*International Classification of Diseases, Ninth Revision*) or K700-K709 (*International Statistical Classification of Diseases and Related Health Problems, Tenth Revision*). Data from the NIS represent hospitalizations rather than individual patients and were weighted across years to calculate national estimates. Primary and secondary diagnosis codes 4215-4217 from the Scientific Registry of Transplant Recipients<sup>4</sup> (an LT database in the US) were used to study AILD-related LTs.

Population estimates were obtained from the US Census Bureau via the CDC WONDER database.<sup>5</sup> The study excluded individuals younger than 20 years and stratified the population into younger (<40 years) and older (≥40 years) patient groups. Joinpoint regression software (version 5.2.0; National Cancer Institute)<sup>6</sup> was used to analyze trends through calculation of mean annual percentage change (MAPC) and SPSS version 29.0.2.0 (IBM) was used for the statistical analysis.

Two-tailed  $P < .05$  was considered significant. The University of Pittsburgh institutional review board approved the study without patient consent requirement.

**Results** | After the data were weighted, 6.5 million AILD-related hospitalizations were distributed among 1.9 million females (28.96%) and 4.6 million males (71.03%) and among 813 506 younger patients (12.57%) and 5.7 million older patients (87.43%). In addition, 27 447 AILD-related LTs were distributed among 6734 females (24.53%) and 20 713 males (75.47%) and among 2800 younger patients (10.20%) and 24 647 older patients (89.80%).

From 2005 to 2021, rates in AILD-related hospitalization (per 100 000 individuals) increased from 63.47 (95% CI, 63.00-63.94) to 129.24 (95% CI, 128.62-129.87) among females, from 186.84 (95% CI, 186.01-187.68) to 301.72 (95% CI, 300.74-302.70) among males, from 33.89 (95% CI, 33.49-34.29) to 102.27 (95% CI, 101.61-102.94) among younger patients, and from 177.93 (95% CI, 177.21-178.65) to 273.48 (95% CI, 272.67-274.29) among older patients ( $P < .001$  for all trends; **Table**). Rates in AILD-related hospitalization (per 100 000 individuals) increased among younger females from 19.53 (95% CI, 19.10-19.97) to 74.83 (95% CI, 74.02-75.65), younger males from 47.91 (95% CI, 47.24-48.58) to 128.91 (95% CI, 127.86-129.97), older females from 88.62 (95% CI, 87.92-89.32) to 157.09 (95% CI, 156.25-157.94), and older males from 278.89 (95% CI, 277.58-280.21) to 403.41 (95% CI, 401.99-404.85) ( $P < .001$  for all trends; **Figure, A**). The greatest relative increases and MAPC occurred among younger females (relative increase, 283.21% [95% CI, 273.86%-292.82%]; MAPC, 8.93 [95% CI, 8.57-9.39]) and younger males (relative increase, 169.07% [95% CI, 164.75%-173.48%]; MAPC, 6.34 [95% CI, 6.00-6.68]) (**Figure, B**).

Between 2005 and 2021, rates in AILD-related LT (per 1 million individuals) increased from 1.88 (95% CI, 1.64-2.16) to 8.39

**Table. Rates of Hospitalizations and Liver Transplants Associated With Alcohol-Induced Liver Disease in the US for 2005-2021**

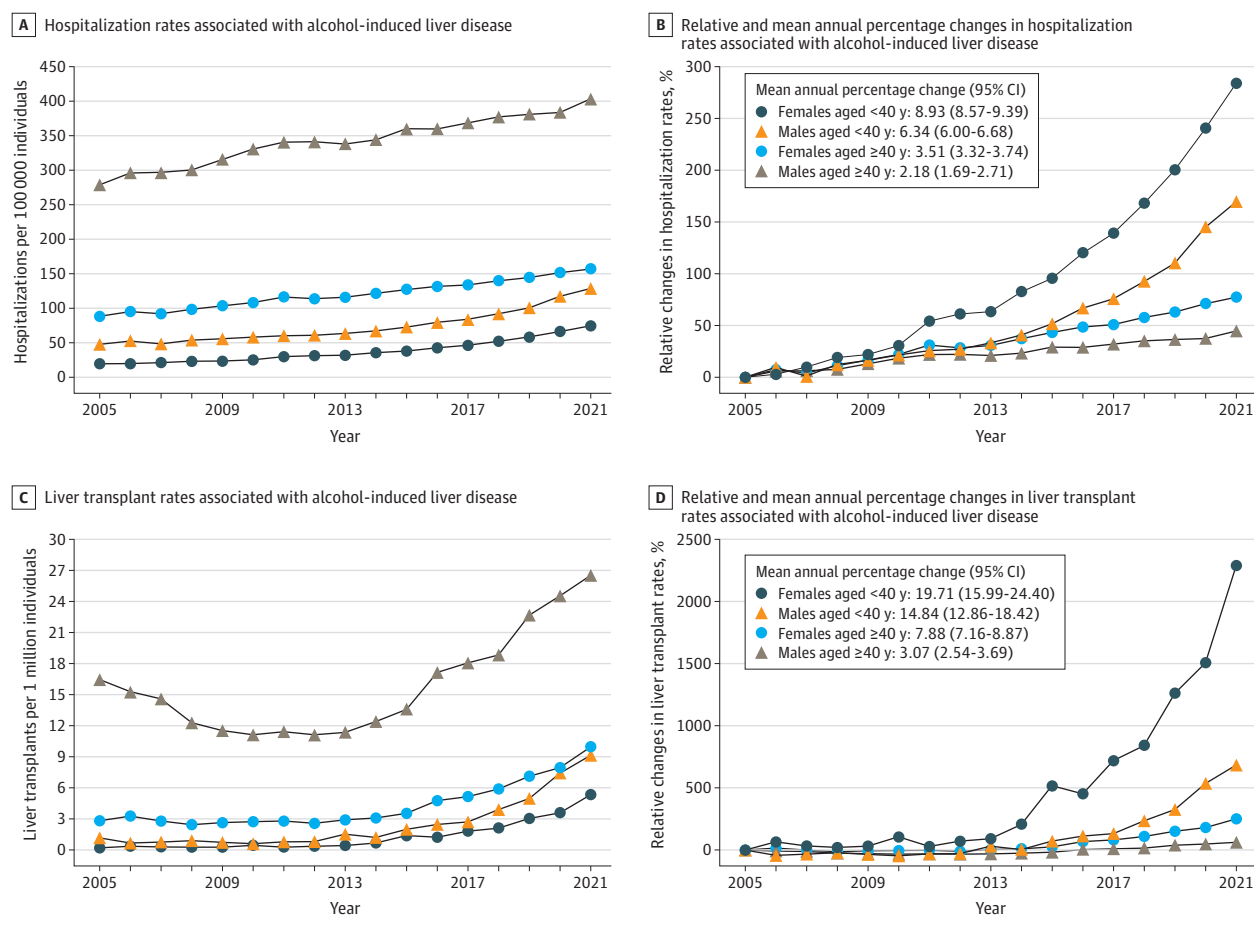
|                               | Rates associated with alcohol-induced liver disease |        |        |        |        |        |        |        |        | RC, %  | MAPC (95% CI)       |
|-------------------------------|-----------------------------------------------------|--------|--------|--------|--------|--------|--------|--------|--------|--------|---------------------|
|                               | 2005                                                | 2007   | 2009   | 2011   | 2013   | 2015   | 2017   | 2019   | 2021   |        |                     |
| Hospitalization <sup>a</sup>  |                                                     |        |        |        |        |        |        |        |        |        |                     |
| Female                        | 63.47                                               | 66.81  | 75.13  | 85.94  | 86.55  | 96.05  | 103.55 | 115.04 | 129.24 | 103.64 | 4.48 (4.16-4.74)    |
| Male                          | 186.84                                              | 199.14 | 214.31 | 232.76 | 233.09 | 250.55 | 260.77 | 275.73 | 301.72 | 61.48  | 2.72 (2.50-2.99)    |
| Age <40 y                     | 33.89                                               | 35.17  | 40.01  | 45.40  | 48.10  | 55.64  | 65.70  | 79.89  | 102.27 | 201.82 | 7.18 (6.91-7.46)    |
| Age ≥40 y                     | 177.93                                              | 188.42 | 203.23 | 221.95 | 220.70 | 237.06 | 244.56 | 256.17 | 273.48 | 53.70  | 2.59 (2.31-2.93)    |
| Liver transplant <sup>b</sup> |                                                     |        |        |        |        |        |        |        |        |        |                     |
| Female                        | 1.88                                                | 1.92   | 1.81   | 1.92   | 2.05   | 2.80   | 4.01   | 5.72   | 8.39   | 345.38 | 9.42 (8.65-10.59)   |
| Male                          | 10.36                                               | 9.16   | 7.34   | 7.34   | 7.60   | 9.17   | 12.25  | 16.05  | 20.09  | 93.84  | 4.21 (3.73-4.84)    |
| Age <40 y                     | 0.70                                                | 0.54   | 0.53   | 0.55   | 0.99   | 1.70   | 2.28   | 4.03   | 7.27   | 937.01 | 16.36 (14.78-19.49) |
| Age ≥40 y                     | 9.23                                                | 8.36   | 6.84   | 6.87   | 6.90   | 8.29   | 11.26  | 14.47  | 17.78  | 92.74  | 4.06 (3.52-4.78)    |

Abbreviations: MAPC, mean annual percentage change; RC, relative change.

<sup>a</sup> Per 100 000 individuals.

<sup>b</sup> Per 1 million individuals.

Figure. Trends in Hospitalizations and Liver Transplants Associated With Alcohol-Induced Liver Disease in the US for 2005-2021



(95% CI, 7.90-8.91) among females, from 10.36 (95% CI, 9.75-11.00) to 20.09 (95% CI, 19.30-20.90) among males, from 0.70 (95% CI, 0.53-0.91) to 7.27 (95% CI, 6.72-7.86) among younger patients, and from 9.23 (95% CI, 8.71-9.76) to 17.78 (95% CI, 17.14-18.45) among older patients ( $P < .001$  for all trends; Table). Rates in AILD-related LT (per 1 million individuals) increased among younger females from 0.22 (95% CI, 0.10-0.43) to 5.34 (95% CI, 4.67-6.07), younger males from 1.17 (95% CI, 0.86-1.55) to 9.15 (95% CI, 8.28-10.08), older females from 2.83 (95% CI, 2.45-3.26) to 9.96 (95% CI, 9.30-10.65), and older males from 16.46 (95% CI, 15.46-17.50) to 26.52 (95% CI, 25.38-27.71) ( $P < .001$  for all trends; Figure, C). The highest relative increases and MAPC occurred among younger females (relative increase, 2283.96% [95% CI, 1134.27%-5179.35%]; MAPC, 19.71 [95% CI, 15.99-24.40]) and younger males (relative increase, 683.54% [95% CI, 480.04%-980.00%]; MAPC, 14.84 [95% CI, 12.86-18.42]; Figure, D).

**Discussion** | This study found significant rate increases for AILD-related hospitalizations and LTs across sex and age groups between 2005 and 2021. Although older males had the highest absolute increases, the greatest relative increases occurred in younger individuals, particularly females. Study limitations include tracking hospitalizations rather than individual pa-

tients, and potential interactions with obesity or viral hepatitis. These findings highlight an important need to elucidate the biopsychosocial factors driving alcohol use and formulate preventive strategies targeted to younger individuals, especially females.

**Prishketan Sethi, MD**  
**Ramon Bataller, MD, PhD**  
**Hao Liu, MD, PhD**  
**Kristine M. Ruppert, DrPH**  
**Michele Molinari, MD, MSc**

**Author Affiliations:** Department of Surgery, University of Pittsburgh Medical Center, Pittsburgh, Pennsylvania (Sethi, Molinari); Liver Unit, Hospital Clínic de Barcelona, Barcelona, Spain (Bataller); Department of Surgery, Houston Methodist Hospital, Houston, Texas (Liu); Department of Epidemiology, University of Pittsburgh School of Public Health, Pittsburgh, Pennsylvania (Ruppert).

**Accepted for Publication:** September 26, 2024.

**Published Online:** November 14, 2024. doi:10.1001/jama.2024.21503

**Corresponding Author:** Michele Molinari, MD, MSc, Division of Abdominal Transplant Surgery, Department of Surgery, UPMC Montefiore, 3459 Fifth Ave, Pittsburgh, PA 15213 (molinarim@upmc.edu).

**Author Contributions:** Drs Sethi and Molinari had full access to all the data in the study and take responsibility for the integrity of the data and the accuracy of the data analysis.

*Concept and design:* Sethi, Bataller, Molinari.

*Acquisition, analysis, or interpretation of data:* Bataller, Liu, Ruppert, Molinari.

*Drafting of the manuscript:* Sethi.

*Critical review of the manuscript for important intellectual content:* Bataller, Liu, Ruppert, Molinari.

*Statistical analysis:* Sethi, Liu, Ruppert, Molinari.

*Administrative, technical, or material support:* Molinari.

*Supervision:* Bataller, Ruppert, Molinari.

**Conflict of Interest Disclosures:** Dr Bataller reported receiving personal fees from Gilead, AbbVie, GSK, Novo Nordisk, and Boehringer Ingelheim. No other disclosures were reported.

**Funding/Support:** This work was supported by grant UL1TR001857 from the National Institutes of Health (awarded to Dr Ruppert and the University of Pittsburgh Clinical and Translational Science Institute).

**Role of the Funder/Sponsor:** The National Institutes of Health had no role in the design and conduct of the study; collection, management, analysis, and interpretation of the data; preparation, review, or approval of the manuscript; and decision to submit the manuscript for publication.

**Data Sharing Statement:** See [Supplement](#).

**Additional Contributions:** We express our sincere appreciation to Oliver Tang, MD (Department of Neurological Surgery, University of Pittsburgh), for his expert feedback and critical review of the data analysis. His contributions significantly enhanced the rigor and clarity of this study. Dr Tang did not receive any compensation for his contribution to the study.

1. Esser MB, Leung G, Sherik A, et al. Estimated deaths attributable to excessive alcohol use among US adults aged 20 to 64 years, 2015 to 2019. *JAMA Netw Open*. 2022;5(11):e2239485. doi:10.1001/jamanetworkopen.2022.39485

2. Cholanteril G, Ahmed A. Alcoholic liver disease replaces hepatitis C virus infection as the leading indication for liver transplantation in the United States. *Clin Gastroenterol Hepatol*. 2018;16(8):1356-1358. doi:10.1016/j.cgh.2017.11.045

3. Agency for Healthcare Research and Quality. National (Nationwide) Inpatient Sample database documentation: 2005-2021. Accessed September 5, 2024. <https://hcup-us.ahrq.gov/db/nation/nis/nisdbdocumentation.jsp>

4. Scientific Registry of Transplant Recipients. Data that drives development: the SRT database. Accessed September 5, 2024. <https://www.srtr.org/about-the-data/the-srtr-database/>

5. US Centers for Disease Control and Prevention. CDC WONDER: state population projections 2004-2030 request. Accessed September 5, 2024. <https://wonder.cdc.gov/population-projections.html>

6. National Cancer Institute. Joinpoint trend analysis software: version 5.2.0. Accessed September 5, 2024. <https://surveillance.cancer.gov/joinpoint/>

## Unionization Efforts by Physicians Between 2000 and 2024

Declining morale and increased corporate employment have led many physicians to consider unionization.<sup>1,2</sup> This study examined trends in unionization among attending physicians (not trainees<sup>3,4</sup>) and the characteristics and motivations for these unions.

**Methods** | We gathered data from the National Labor Relations Board (NLRB) on all petitions to form a union that included physicians from 2000 through May 2024.<sup>5</sup> We collected information on the number of individuals in the petitions, certification (approved by a majority of the eligible members in an election), membership (physicians and nonphysicians) of the bargaining unit, location, and employer.<sup>6</sup>

Because filings appeared to increase in 2023, we compared petitions filed between 2000 and 2022 with those filed from 2023 through May 2024. We calculated descriptive sta-

tistics on petition data and annual rates (with 95% CIs calculated using the Poisson exact method) of filings for these periods using R version 4.2.1. We identified statistical significance as the lack of overlap of the 95% CIs for annual filing rates for the 2 periods. This study was nonhuman subjects research obviating the need for institutional review board approval.

We also qualitatively assessed motivations for unionization by surveying press reports for the unionization drives in 2023-2024 using Google searches including the location where the union drive was occurring, the hospital name, and the terms *physician* and *union*.

**Results** | There were 77 union petitions filed with physician members (44 in 2000-2022 and 33 in 2023-2024), representing 7064 individuals (3541 in 2000-2022 and 3523 in 2023-2024; **Figure**). The annual rate of union petition filings was 3.2 (95% CI, 2.5-3.9). It was 2.1 (95% CI, 1.5-2.8) in 2000-2022 and 23.3 (95% CI, 16.0-32.7) in 2023-2024. Overall, excluding petitions that had yet to have a disposition, 41 of 66 (62%) were certified: 24 of 44 (54%) in 2000-2022 and 17 of 22 (77%) in 2023-2024.

The size of the bargaining unit ranged from 4 to 560 (mean, 93 [SD, 116]; median, 45 [IQR, 23.5-140]) employees; 34% of the unions were exclusively physicians and doctors of osteopathy, 40% were composed of physicians and advanced practice clinicians (eg, nurse practitioners and physician assistants), and 26% represented a broader coalition, including physicians, nurses, and administrative personnel. Petitions were concentrated in western states, with 43 of the 77 in California, Oregon, and Washington.

Petitions were filed against a diversity of employers: 49% were filed against hospitals, 38% against community health centers, and 13% against nonhospital corporate owners, such as Optum and private-equity-backed companies.

Of the 33 petitions in 2023-2024, we found press reports documenting the motivations for the union campaigns for 26 (79%). Union campaigns were primarily motivated by non-compensation concerns: 85% cited working conditions, 81% lack of voice in management, 54% patient care concerns, whereas only 1 (4%) cited financial compensation.

**Discussion** | Compared with 2000-2022, the number of union petitions with physicians in the bargaining units filed and certified increased in 2023 through May 2024. Organizing efforts were motivated by concerns about working conditions, physicians' autonomy, and voice in management, and the quality of patient care.

Limitations include that public-sector physician unions governed by state law and unions voluntarily recognized by the employer were not included. For bargaining units with a diverse range of employees, physicians may only comprise a subset of the individuals in the unit. For the assessment of motivations, statements made to news outlets may reflect public relations determinations. Although duplicates of the same bargaining unit were screened out, individual petitions against the same employer from different bargaining units were counted as unique filings.